

Matthew N. Czuba • www.czu.ba • czuba@ucla.edu • <https://linkedin.com/in/matthewczuba> • **Citizenship:** USA

Interests: innovation in generative AI, LLMs, natural language syntax/semantics, AI-driven NLP & parsing systems

Career Statement: Passionate, highly driven technologist and skilled AI researcher with a unique blend of theoretical depth, systems engineering expertise, and commercial intelligence. My early foundation in theoretical linguistics (9 publications across 6 languages) directly informs hands-on work engineering production NLP systems at Melange, where I architect and maintain our core patent processing pipeline – implementing custom CYK parsers, context-free grammars, entity graph algorithms, and distributed LLM prompting infrastructure deployed on GCP. This technical breadth is enhanced by my proven GTM record at Speechmatics (160% of target), where I served as a strategic technical liaison bridging Product, Marketing, and Finance. My unique value proposition lies not only in translating bleeding-edge CS theory (formal grammars, graph algorithms, computational linguistics) into production-grade enterprise systems, but also in the articulation of technical capabilities into compelling business narratives that drive adoption. I deliver complete product lifecycles from research conception through scalable cloud deployment with measurable ROI.

Melange Technologies • NLP Systems Engineer & Head of Partnerships • *October 2024–present*

- **Customer Solutions Engineering:** Serve as the primary technical partner for enterprise customers on discovery and scoping calls. Translate client search requirements into concrete NLP pipeline designs & workflow, owning implementation, validation, and iteration through to production use, ensuring delivered behavior matches agreed success criteria.
- **Cross-Functional GTM/Product:** Partner in strategic Product & GTM initiatives to translate NLP insights into practical featurization for PMF. Collaborate closely with founders, business development, and product to shape solution scope, communicate technical trade-offs to non-technical stakeholders, and feed customer requirements back into roadmap.
- **AI System Architecture:** Designed and implemented core components of Melange's NLP system pipeline, creating production-quality NLP systems using theoretically grounded context-free grammars and dynamic-programming algorithms.
- **Production Ownership:** Core contributor to enterprise patent NLP pipeline processing 50+ patents quarterly; own all client-facing production operations including bugfixes, client patent processing, and deliverable-based incident resolution.
- **Advanced NLP Engineering:** Architected and implemented CYK bottom-up parser with automatic Chomsky Normal Form conversion and custom context-free grammar notation (optionality, disjunction, Kleene star), achieving $O(n^3)$ complexity for patent claim parsing across custom context-free grammars tailored to complex patent claim syntactic parsing tasks.
- **Hybrid Symbolic-Neural Systems:** Designed integration of formal grammars with LLM prompting for syntactic ambiguity resolution (constituent, conjunction, attachment), building and maintaining 20+ structured prompt templates for entity resolution, antecedent identification, and parse tree disambiguation, with automatic retry and cache invalidation.
- **Entity Graph Algorithms:** Engineered core entity graph data structures for cross-claim dependency resolution and coreference tracking, enabling robust component-to-limitation linkage across hierarchical patent claim structures.
- **Cloud Infrastructure & Performance:** Designed multi-layer Redis caching and deployed on GCP (Docker, Cloud Run, Cloud Build), incorporating grammar content and subphrase dependencies for deterministic cache invalidation across distributed Cloud Run instances and achieving ~75% cache hit rates and $\sim 3\times$ throughput improvements on patent workloads.

Speechmatics USA • Strategic Business Development • *February 2024–October 2024*

- **Business Development:** Consistently exceeded all Business Development KPIs, achieving 160% of lead conversion quota, generating \$1M in pipeline revenue through strategic prospecting, exceptional outreach, technical demos, & brand evangelism.
- **Partnerships:** Fostered partnerships with unicorn logos from cold outreach and on-the-ground relationship-building.
- **Cross-functional:** Served as the sole Business Development liaison for the cross-functional Product team involved in strategically increasing usage and demand for specific non-English Language initiatives, maximizing the effectiveness of individual model uplifts and landings; also as the representative Business Development liaison between Product Marketing and Finance to develop internal Competitive Intelligence and drive GTM outreach initiatives based on expert market analysis.
- **RevOps & Marketing:** Drove improvements to marketing & revenue operations, streamlining the efficacy of MQLs by highlighting inefficiencies in the data collection of web portal user input, and writing technically engaging public-facing copy.

UCLA Department of Linguistics • Graduate Ph.D Research Scientist • 2019–September 2023

- **Research Vision:** Owned an individual research agenda deploying the latest in theoretical linguistics across 6 languages.
- **Financial Success:** Exceeded all KPIs in negotiating >200% of competitive funding quota for individual and team projects.
- **Language Model Design & Development:** Hands-on computational linguistic R&D experience developing FSGs, weighted CFGs, and other non-ANN models across all domains of natural language in Haskell and Python.
- **Data Collection:** Designed and oversaw elicitation sessions extracting 300+ hours of language data from 12 participants.
- **Data Annotation:** Hands-on experience with data annotation (Praat) and analysis to aid in the development of theoretical phonological, syntactic, semantic, and pragmatic models of natural language and communication.
- **Statistical:** Hands-on experience performing statistical analyses of experimental data for linguistic research using R.
- **Innovation & Knowledge Sharing:** Regularly collaborated to produce novel discoveries in natural language science.
- **Presentation & Publication:** Contributed to team success authoring 13 conference presentations, publishing 9 papers.
- **Global Representation:** Traveled to 5 countries in Europe/Asia delivering technical presentations on behalf of the team.
- **Industry Event Organization:** Led conferences with 300+ attendees, multitasking speaker selection, website design, social media, and team management, building and maintaining strong relationships across international universities.
- **Technology Adoption:** Stayed abreast of the latest developments in technological trends, producing research summaries for departmental strategic planning, sharing knowledge, and driving innovation independently and as a key team player.

Harvard Journal of East Asian Linguistics (JEAL) • Invited Reviewer • 2022-2023

- **Expert Authority:** Invited to perform peer review of journal submissions in the areas of Chinese theoretical syntax.

USC North American Summer School in Language, Logic, and Information • Career Mentor • 2022

- **Dedicated Mentorship:** Invited to consult, providing valuable one-on-one guidance, expert career training, and mentorship services, nurturing the educational ambitions of aspiring undergraduate researchers in the natural language sciences.

UCLA Department of Linguistics • Primary Instructor and Teaching Assistant • 2020-2022

- **Exceptional Student success:** Achieved a 98% pass rate as the primary instructor for UCLA's undergraduate linguistics.
- **Engaging Curriculum:** Designed and delivered comprehensive coursework for over 750 students, fostering excellency.
- **Effective Leadership:** Navigated academic logistics by bridging communication between admin, faculty, and students.
- **Team Management:** Led a team of assistants, ensuring their support and effective contribution to the learning process.

Education

- M.A., Generative Linguistics, 2023. University of California, Los Angeles (UCLA) Department of Linguistics.
- Ph.D coursework in mathematics, computer science, cognitive science, psychology, formal logic, & NLP.
- M.A., Theoretical Linguistics, 2019. Institute of Linguistics, National Tsing Hua University (NTHU).
- B.A., Cognitive Science and Mandarin Chinese, 2016. Marlboro College. Graduated with Highest Honors.

Proficiencies & Technical Experience

- *Programming & Version Control:* Python, Haskell, Lisp, Go, fish, zsh, Bash, Git, JavaScript, R, LaTeX
- *NLP Tools:* Stanza, spaCy, NLTK, Praat, ELAN, Jupyter
- *AI Integration:* LLM APIs (OpenAI, Anthropic), Cursor AI (VS Code), emacs + aidermacs
- *Cloud & APIs:* GCP (Cloud Run/Build), REST APIs, OAuth, webhooks, FastAPI, Redis, Docker, CI/CD
- *Data:* Linguistic Corpus Development, Data Annotation, Statistical Analysis, Experimental Research Design
- *Business Intelligence:* Technical Demo Design, CRM (Salesforce), Product Positioning, GTM Strategy
- *Languages:* Polish (fluent, heritage), Mandarin (fluent)
- *Soft:* Cross-functional Collaboration, Technical Documentation, Research Planning, Technical Presentations